

Module 5: Supply Chain Management

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Supply Chain Management (SCM) in E-Commerce

Supply Chain Management (SCM) refers to the systematic coordination and management of all activities involved in sourcing, procurement, production, inventory management, transportation, warehousing, and delivery of goods and services from suppliers to final customers.

In e-commerce, SCM plays a critical role because online businesses depend heavily on efficient movement of products, accurate information flow, and timely delivery services. A well-managed supply chain reduces operational costs, improves customer satisfaction, and increases overall business efficiency. SCM integrates various entities such as:

- Suppliers
- Warehouses
- Retailers
- Manufacturers
- Distributors
- Customers

using internet technologies, enterprise systems, and digital communication networks.

1) E-Logistics

E-Logistics refers to the application of internet technologies, electronic communication systems, and digital platforms in logistics and transportation management. It involves the electronic handling of activities such as inventory management, shipment tracking, warehousing, transportation, and order fulfilment.

E-logistics is one of the most important components of e-commerce because online businesses require fast, accurate, and efficient delivery systems.

Functions of E-Logistics

- 1) Online Order Processing:** Customer orders are automatically processed through digital systems without manual intervention.
- 2) Inventory Management:** Tracks stock levels in real time and updates inventory databases automatically.
- 3) Shipment Tracking:** Customers and businesses can monitor product movement and delivery status online.
- 4) Warehouse Coordination:** Manages storage, packaging, and movement of goods within warehouses.
- 5) Transportation Management:** Optimizes delivery routes and shipping schedules.

Features of E-Logistics

- Real-time tracking systems

- Automated inventory control
- Electronic documentation
- Integration with ERP and SCM systems
- Faster communication between supply chain partners

- Better inventory accuracy
- Enhanced coordination among supply chain partners
- Reduction in paperwork and manual processes

Advantages of E-Logistics

- Faster delivery and reduced lead time
- Lower transportation and operational cost
- Improved customer satisfaction

Limitations of E-Logistics

- High initial setup cost
- Dependence on internet infrastructure
- Cybersecurity and data privacy risks
- Technical failures may disrupt operations

2) Supply Chain Portal

A Supply Chain Portal is a centralized web-based platform that allows different supply chain participants such as suppliers, manufacturers, distributors, logistics providers, and customers to access, share, and manage supply chain information electronically.

It acts as an integrated communication and collaboration system for supply chain activities.

Functions of Supply Chain Portal

- 1) Information Sharing:** Provides real-time access to inventory, order status, and shipment details.
- 2) Supplier Coordination:** Facilitates communication and collaboration between suppliers and businesses.

- 3) Procurement Management:** Supports online purchasing, supplier selection, and procurement activities.
- 4) Delivery Tracking:** Allows tracking of goods from warehouses to customers.
- 5) Document Management:** Stores invoices, purchase orders, contracts, and shipping documents electronically.

Advantages of Supply Chain Portal

- Improved transparency across the supply chain
- Faster communication and decision-making

- Better coordination among business partners
- Reduced operational delays
- Increased efficiency and productivity

Importance in E-Commerce: In e-commerce, supply chain portals help businesses manage large numbers of transactions and coordinate with global suppliers efficiently.

3) Supply Chain Planning Tools (SCP Tools)

Supply Chain Planning (SCP) tools are software applications used for forecasting, planning, and optimizing supply chain operations. These tools help organizations make strategic decisions regarding inventory, production, transportation, and demand management.

Functions of SCP Tools

- 1) **Demand Forecasting:** Predicts future customer demand using historical sales data and analytics.
- 2) **Inventory Planning:** Determines optimal inventory levels to avoid shortages or overstocking.
- 3) **Production Planning:** Schedules manufacturing activities based on market demand.
- 4) **Resource Allocation:** Allocates labour, machines, and materials efficiently.
- 5) **Distribution Planning:** Optimizes product distribution and transportation strategies.

Advantages of SCP Tools

- Reduces inventory holding cost
- Improves forecasting accuracy
- Minimizes supply chain disruptions
- Enhances resource utilization
- Improves customer service and product availability

Examples of SCP Activities

- Sales forecasting
- Capacity planning
- Procurement planning
- Transportation scheduling

4) Supply Chain Execution (SCE)

Supply Chain Execution (SCE) refers to the management and execution of day-to-day supply chain operations. It focuses on the actual movement, storage, and delivery of products throughout the supply chain.

While SCP focuses on planning, SCE focuses on implementation and operational activities.

Functions of SCE

- 1) **Order Management:** Processes customer orders and coordinates fulfilment activities.
- 2) **Warehouse Management:** Controls storage, packaging, picking, and movement of products inside warehouses.
- 3) **Transportation Management:** Manages shipping operations, routing, and carrier selection.
- 4) **Inventory Tracking:** Tracks stock movement and product availability in real time.
- 6) **Delivery Coordination:** Ensures timely delivery of products to customers.

Objectives of SCE

- Reduce delivery delays
- Improve operational efficiency
- Minimize logistics costs
- Enhance customer satisfaction
- Ensure accurate order fulfilment

5) SCE Framework

The SCE Framework is the structural model that integrates various execution activities and systems involved in supply chain operations.

It ensures smooth coordination between inventory, warehousing, transportation, and customer service systems.

Components of SCE Framework

- 1) **Order Management System:** Handles order entry, processing, tracking, and customer communication.
- 2) **Warehouse Management System (WMS):** Controls warehouse operations such as:
 - Storage allocation
 - Inventory control
 - Picking and packing
- 3) **Transportation Management System (TMS):** Manages:
 - Route planning
 - Carrier management
 - Shipment scheduling
 - Delivery tracking
- 4) **Inventory Management System:** Monitors stock levels and product availability across supply chain networks.
- 5) **Information and Communication System:** Facilitates real-time data exchange between suppliers, warehouses, logistics providers, and customers.
- 6) **Internet's Effect on Supply Chain Power:** The internet has transformed traditional supply chains into highly connected digital networks. It has changed the balance of power among suppliers, businesses, and customers.

Positive Effects of Internet on Supply Chain

1. **Increased Transparency:** Businesses and customers can access real-time information about products, inventory, and shipments.
2. **Faster Communication:** Internet-based systems allow instant communication between all supply chain participants.
3. **Improved Collaboration:** Suppliers, manufacturers, and distributors can coordinate activities more efficiently using online platforms.
4. **Customer Empowerment:** Customers can compare prices, products, and reviews easily, increasing their influence in the market.
5. **Disintermediation:** The internet allows manufacturers to sell directly to customers, reducing dependence on intermediaries.
6. **Global Market Access:** Businesses can connect with international suppliers and customers through digital networks.

Challenges Caused by Internet Integration

- Increased competition in digital markets
- Cybersecurity threats and data breaches
- Pressure for faster delivery and service quality
- Dependence on internet connectivity and technology infrastructure